

বর্ণমালা · Bornomala

An offline Bangla SMS app with three jobs: compress ordinary messages so 5× more fits in one text, compose emergency reports any phone can read, and let one volunteer relay many households' reports in a single SMS.

Everything runs on the phone. No server, no internet, no account, no message ever leaves the device except through your own SMS app.

mayamoho.github.io/bornomala · github.com/Mayamoho/bornomala

In July 2024 the state cut the mobile internet.

Broadband went dark too. What kept working was the oldest part of the phone network: voice calls and text messages. People fell back to texting.

Every crisis tool assumes it can build a new network. In July, nobody could.

And it costs twice as much to text in Bangla.

Text messages were built around an old character set that only covers English. Write one Bangla letter and the whole message switches to a heavier format:

160

characters per segment, English

70

characters per segment, Bangla

2.3×

more expensive, per message

More segments means more cost, more retries, and more failures on a congested network — exactly when SMS is the only channel left.

A mesh reaches the room. A text reaches your mother.

Half of crisis tech is offline mesh messengers — Bluetooth or Wi-Fi passing messages phone to phone. Good idea, three conditions attached:

	BLUETOOTH MESH APP	BORNOMALA
Who needs the app	both people, installed <i>before</i> the disaster	only the sender
How far it reaches	about 100 metres	the whole country
Who it can reach	strangers nearby who also installed it	any phone that can receive a text
Can it call 999	no	yes, one tap

So Bornomala does not build a network. It uses the one that already reaches everybody — text messages and voice calls — and makes it cheaper.

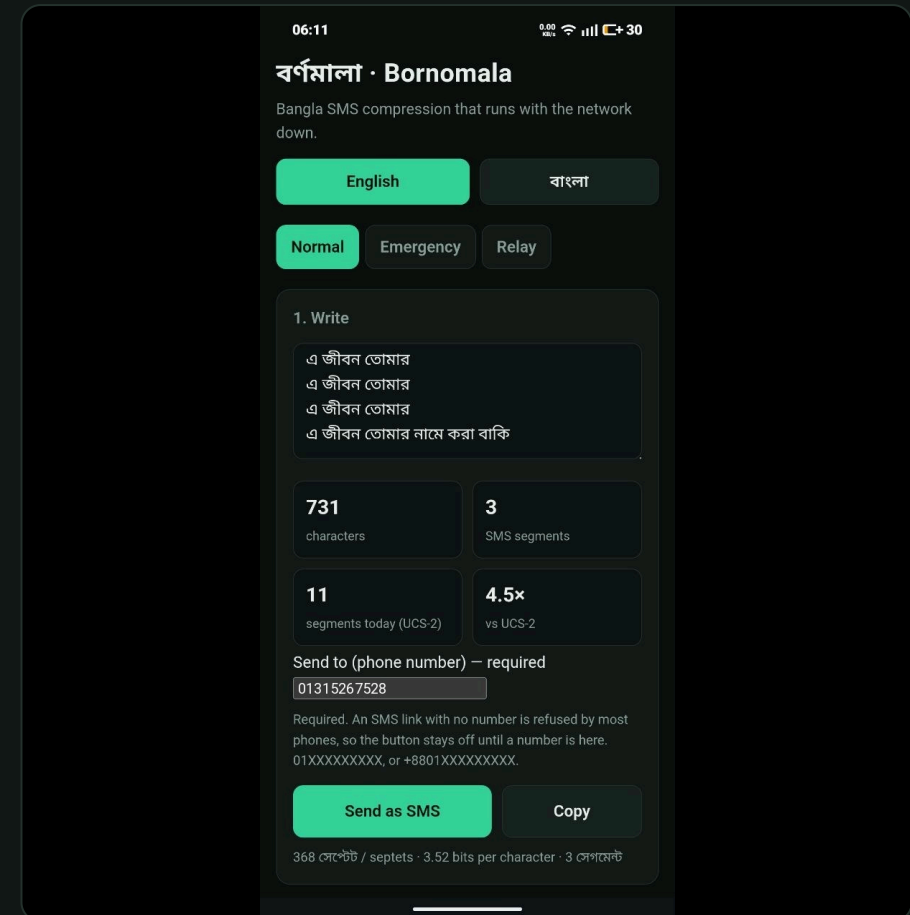
It guesses the next letter, so the letter costs less.

- 1 A small Bangla model rides inside the app. No internet, no server.
- 2 It predicts each next letter. A good guess costs almost nothing to record.
- 3 The bits go out as characters SMS treats as cheap.

আমরা পাঁচজন নিরাপদ আছি, ঢাকা

Bù8P9*+=! ¤1+1j

731 characters → 3 texts, not 11

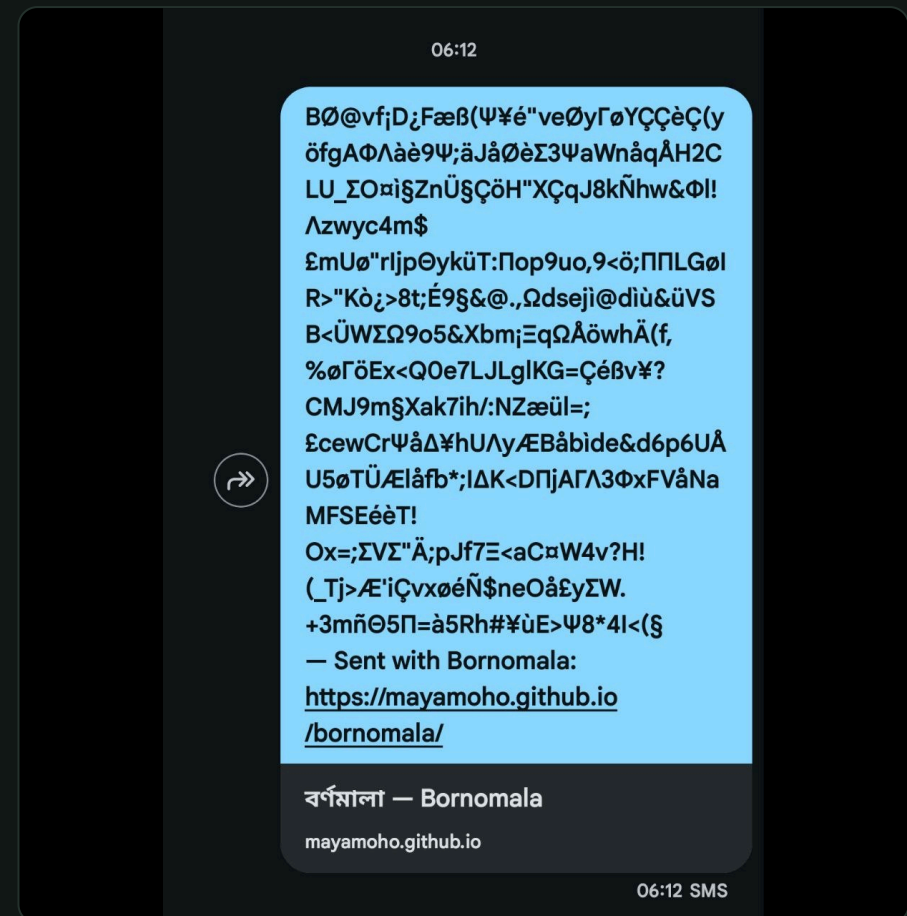


731 Bangla characters: 3 texts, not 11.

Type Bangla. Send it. Read it back.

- **Send** opens your own SMS app, number and message filled in.
- It stays off until a real number is there — a text link with no number silently fails.
- ← Paste the whole message you received. It finds the code, ignores the signature and the map link.
- Anything shared into the app from elsewhere lands in the decode box.

Try it: 01315267528



How it lands on the other phone.

A survivor has no app. So this tab sends words.

- What is happening, in your own words.
- One of 32 ready sentences, with a blank to fill.
- A district, or your live location.
- How long ago.
- 999 and eight more, one tap to call — these are call centres, so calling is the reliable path. **Calls need no internet.**

It refuses to send until those are answered — a report with no place is not a report.

06:17

English বাংলা

Normal Emergency Relay

1. Your message *

Nothing here is compressed or coded. Anyone can read it — the receiver needs no app.

বন্যায় একটি বাড়িতে ৭ জন আটকা পড়ে আছি, শিশু ও মহিলা আছে, বাড়ি অর্ধেক পানির নিচে ভলিয়ে গেছে, সাহায্য প্রয়োজন।

2. Add a ready sentence *

Tap one, or pick from the list.

☒ of us are safe SOS need help ()
☐ people trapped injured ()
drinking water needed for ____ evacuate this area ()
Or pick from the list
Choose...
3. Where are you *
Choose...
4. When *
Choose...

Typing the report.

06:19

5. What gets sent

Send to (phone number) — required

01315267528

Required. An SMS link with no number is refused by most phones, so the button stays off until a number is here. 01XXXXXXXXXX or +8801XXXXXXXXXX.

বন্যায় একটি বাড়িতে ৭ জন আটকা পড়ে আছি, শিশু ও মহিলা আছে, বাড়ি অর্ধেক পানির নিচে ভলিয়ে গেছে, সাহায্য প্রয়োজন।

7 people trapped : just now
23.79319, 90.37090
https://maps.google.com/?q=23.79319,90.37090
— Sent with Bornomala:
https://mayamoho.github.io/bornomala/

262 characters 4 SMS segments

Send as SMS Copy

Add to relay

Plain text — readable by anyone, no app needed.
On a computer nothing will happen — SMS needs a phone. Use Copy instead.

Show QR

No network at all? Show this to the phone beside you and

What actually gets sent — plain words.

06:22

6. Emergency numbers

Tap to call, or send this message to them as an SMS. 999 is toll-free. Short codes do not always accept SMS — if it fails, call.

999 — National emergency — police, fire, ambulance Call SMS
1090 — Disaster warning Call SMS
102 — Fire Service & Civil Defence Call SMS
16263 — DGHS health helpline Call SMS
333 — Government information & services Call SMS
109 — Violence against women and children Call SMS
1098 — Child helpline Call SMS
16430 — National legal aid Call SMS

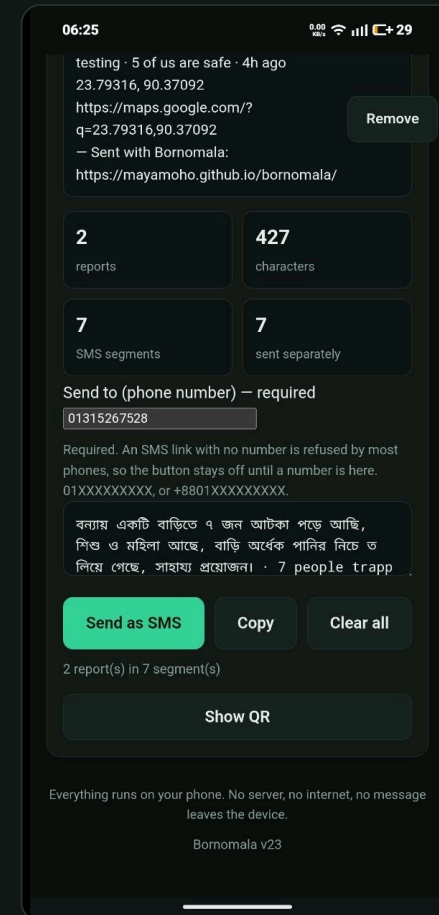
Nine national numbers.

Forty families. One message.

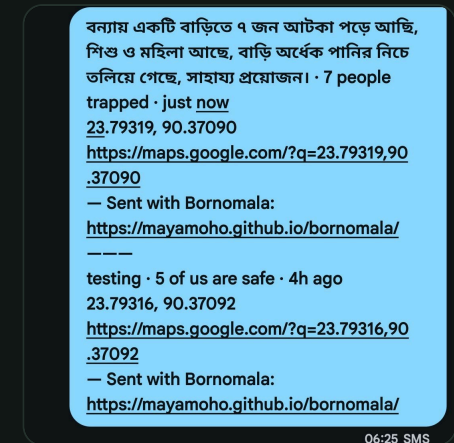
- + Each household is added with one tap from the Emergency tab.
- = They go out together, with one signature for the batch.
- △ A counter shows what sending them one by one would have cost.

Two households → one message

When the towers are jammed, one message that gets through beats forty in a queue.



The queue, and what it costs.



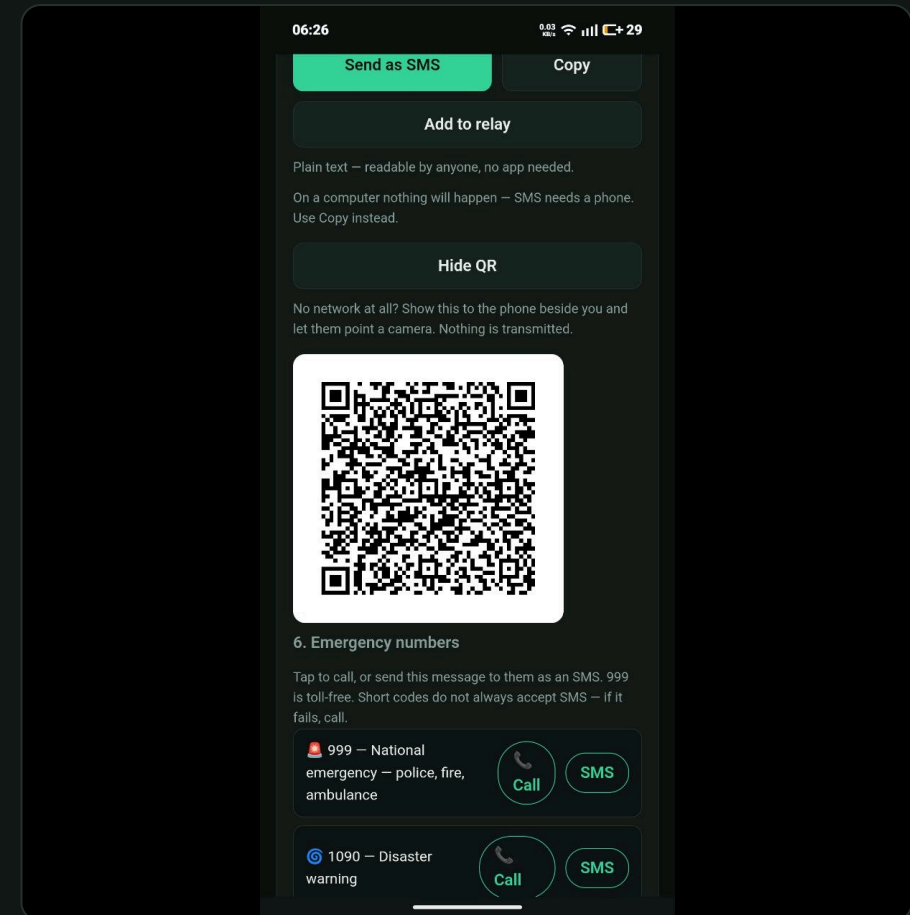
Both arriving as one SMS.

5,000 messages the model had never seen.

	BANGLA IN ONE TEXT	FITS IN ONE
Bornomala	361	100.0%
Every phone today	70	98.2%
gzip	53	99.5%

5.15× more Bangla per text

Ordinary compression does worse than doing nothing — it has no idea what Bangla looks like, and a message this short never repays the overhead.



No network at all? The message becomes a QR.

Each layer below has a floor.

No internet

install it once and it opens forever after, with no connection

No mobile data

texts and calls still go, over 2G

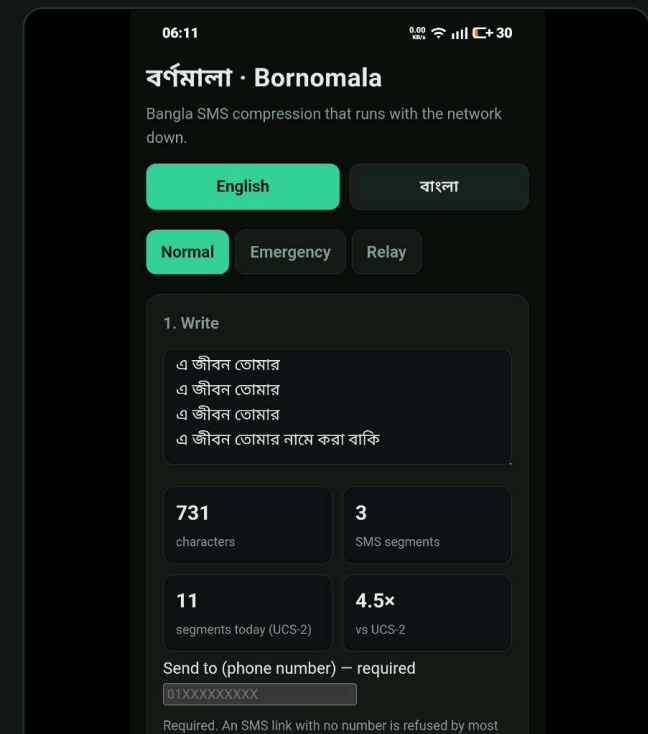
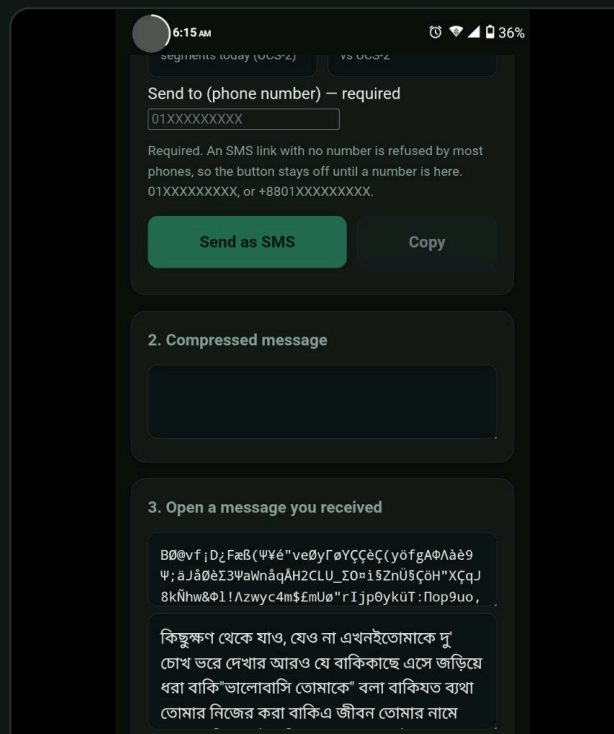
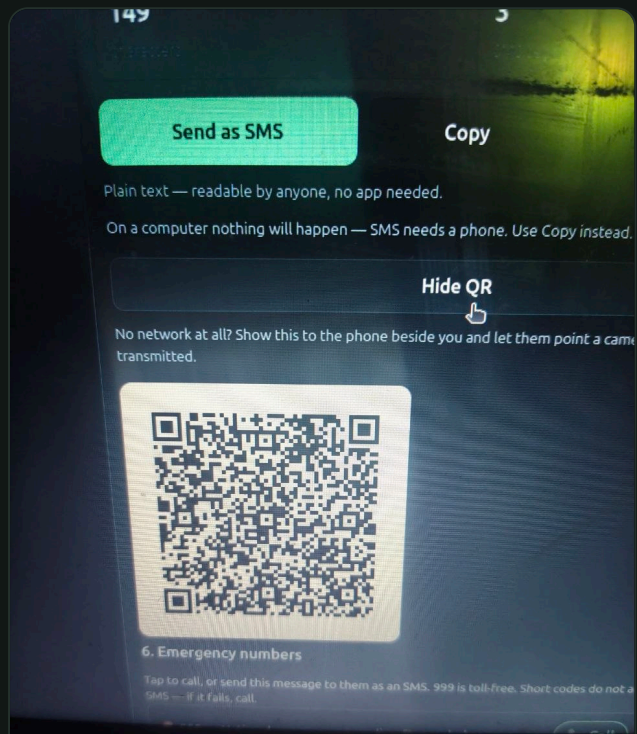
No GPS

pick your district instead of using live location

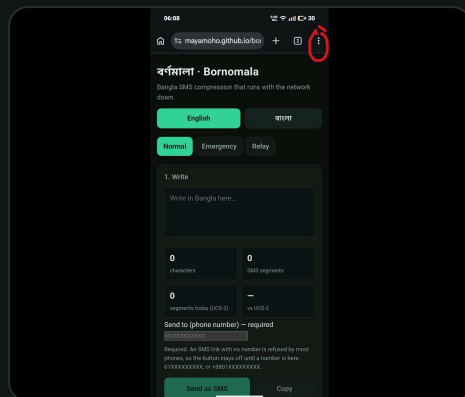
No network at all

show the QR code and let the phone beside you photograph it

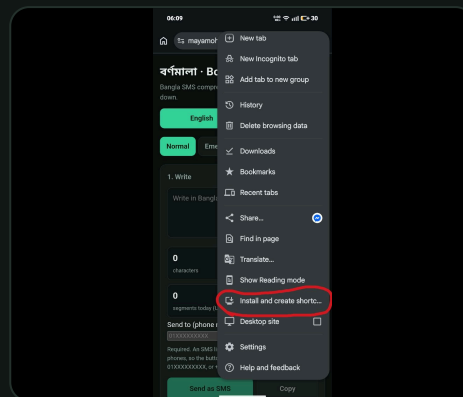
Your location always travels twice — as a map link *and* as plain numbers beside it — because the numbers still make sense on a phone that cannot open the link.



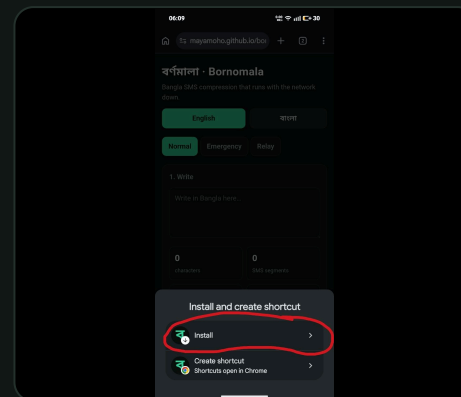
Installed, sent, received — on two Android handsets.



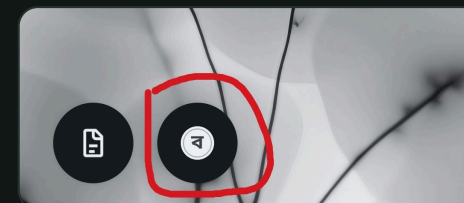
Chrome's : menu.



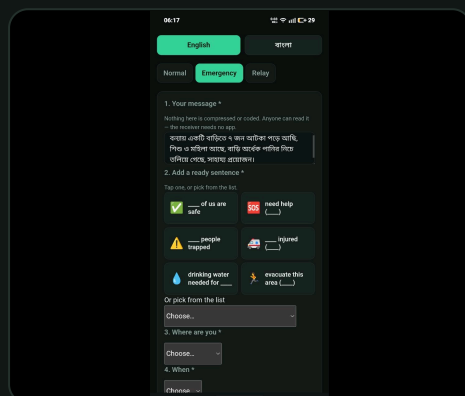
Install and create shortcut.



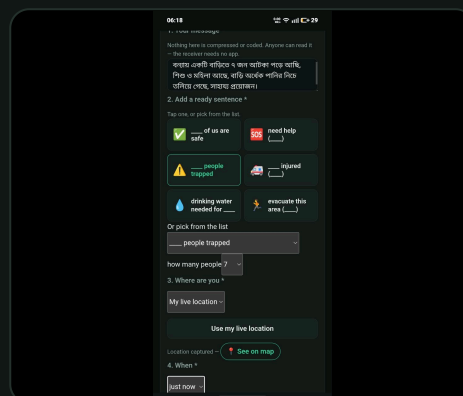
Confirm Install.



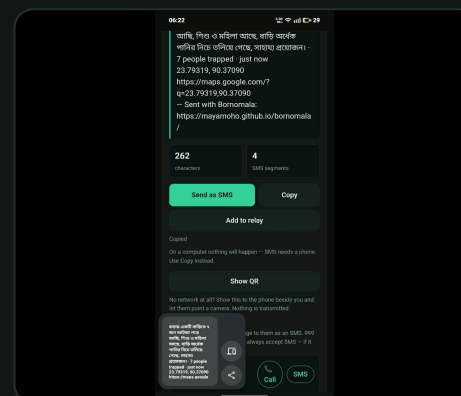
On the home screen.



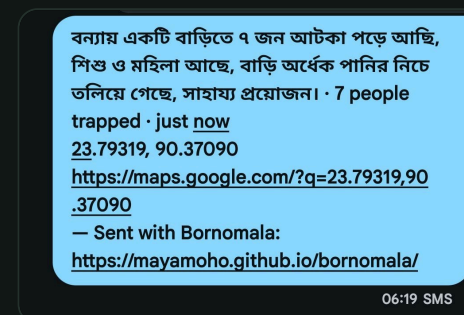
Typing the report.



Live location captured.



Copy and share out.



Arriving as plain words.

Live at mayamoho.github.io/bornomala · bilingual, switchable mid-message · 81 automated checks across three suites · MIT licensed, with the benchmark and the model training tools in the repo.